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Control Valve Diagnostic Data Acquisition and Reporting

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Control Valve Diagnostic Data Acquisition and Reporting

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Foreword

A need exists for a guide for acquiring and reporting of diagnostic data when using automated valve data acquisition devices. These are usually externally applied, portable data acquisition devices or microprocessor based positioners. However, the provisions of this document may be applied to any method of data acquisition as long as it meets the quality, type, and frequency of data recorded.

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1 Scope

This document applies to all pneumatically operated, automated rotary or reciprocating, on/off, or modulating valves. It also includes automation components (i.e. positioners, transducers, and solenoids) as applicable. It provides a methodology for standardizing the acquisition and reporting of data used in assessing valve condition. The document includes the type of data to be acquired, the number of measurements to be recorded and their resolution, units of measure, nomenclature, computer file storage format, graphical presentation, and other reporting formats. It does not address interpretation of data or diagnosis of valve condition.

All data reported per this document are acquired with the valve out of service with no internal dynamic forces acting on the closure member.

2 Purpose

2.1 The purpose of this document is to provide users of diagnostic products with a uniform means of acquiring and reporting data used for diagnosing valve operability. This is necessary to provide the end user with a means of easily and accurately comparing diagnostic results from acquisition devices made by various manufacturers.

2.2 The following is intended to provide minimum resolution and definition of diagnostic tests performed. It does not preclude performing testing with more definition or resolution, or doing other testing with which to diagnose valve performance.

3 Definitions

3.1 actuator differential pressure:
the pressure differential across the piston(s) on double acting cylinders.

3.2 actuator pressure(s):
the input pressure(s) to the actuator.

3.3 valve position versus actuator pressure signature:
a plot of valve position versus actuator pressure (differential pressure for double acting) in both directions, over the minimum to maximum signal pressure specified for the application.

This signature may be generated by varying the control signal to a positioner or a pressure signal to the actuator. However, it must be clearly stated on the graph and in the other reported data which method was used.

3.4 diagnostic test data:
the set of valve operational data collected. This data is taken over the travel specified for the signature and may be reported graphically, as a tabulation of data points, or as a single element. The following is a list of typical diagnostic test data.

a) Input signal

b) I/P transducer output

c) Positioner output pressure (see note)

d) Actuator signal pressure(s) (see note)

e) Valve position

f) Supply pressure

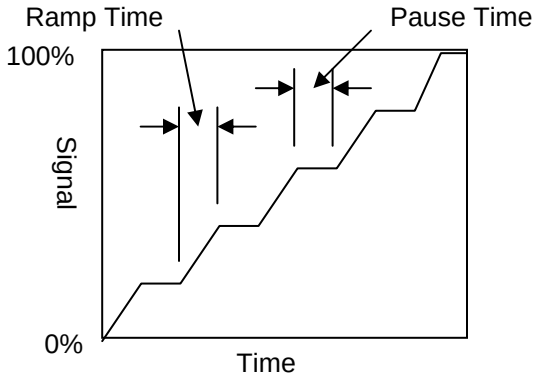
g) Solenoid (energized or de-energized)

NOTE — The actuator pressure, transducer output pressure, or positioner output pressure may be assumed to be the same if the transducer (in cases where no positioner is used) output or electro-pneumatic positioner output is connected directly to the actuator or by an unobstructed conduit.

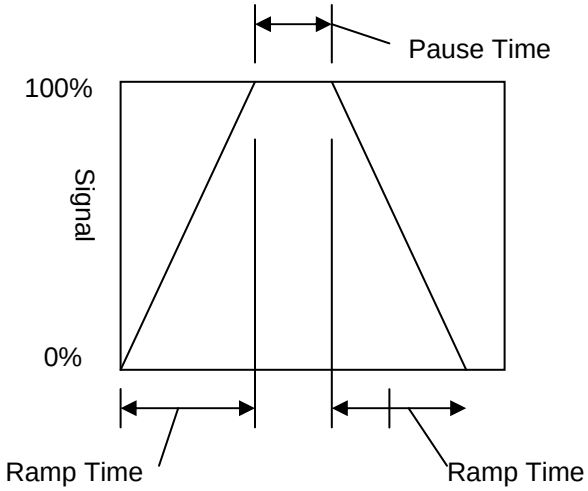
3.5 dynamic signature:
the signature produced by operating a valve through its signal range while acquiring data points throughout the range.

3.6 pause time:
the period of time in a signature from the end of the input signal increase or decrease to the beginning of a new input signal increase or decrease. The duration of pause time must be enough for the valve to stabilize. See illustrations in “ramp time” definition.

3.7 ramp time:
for the ramp and pause test method, it is the time allowed for the valve to achieve a new signal position after a change in input signal. For the dynamic test method it is the total time to ramp the signal from the user-specified minimum to the user-specified maximum at a constant rate.



Ramp and Pause Method



Dynamic Test Method

3.8 valve position:
the actual position of the valve closure member relative to the valve rated travel.

3.9 valve travel:
see ANSI/ISA-75.05.01-2000 (R2005).

3.10 step and pause signatures:
the signature produced by signaling the valve to move a portion of the span over which the signature will be produced, pausing to allow valve stabilization, recording a data point, and repeating this process until the signature is determined.

3.11 transducer output versus control signal signature:
a plot of I/P transducer output signal pressure versus I/P transducer input signal.

3.12 rated travel:
see ANSI/ISA-75.05.01-2000 (R2005).

3.13 input signal:
the signal directly applied to the device to generate a signature.

3.14 control signal:
the specified controller input signal regardless of how it is generated.

3.15 supply pressure versus control signal signature:
a plot of the control signal versus the supply pressure over one full travel cycle.

3.16 valve position versus control signal signature:
a plot of valve position opening and closing versus positioner input signal.

4 Diagnostic test standard

4.1 Data will be acquired at the number of equally spaced intervals of input signal specified over the travel in the opening direction and the closing direction. The recommended minimum number of data points in each direction is 101 for a full signal span.

4.2 Data will be acquired using the following process.

a) Ramp and pause test method

1) User defines signal range over which the test is performed.

2) Starting at the user defined minimum signal specified, generate 1% or less change in total valve control signal (input) span.

3) Wait for the ramp and pause time specified.

4) Record the following data.

a) Input signal

b) I/P transducer output, if used

c) Positioner output pressure(s) (see note)

NOTE — The actuator pressure, transducer output pressure, or positioner output pressure may be assumed the same if the transducer (in cases where no positioner is used) output or electro-pneumatic positioner output is connected directly to the actuator or by an unobstructed conduit.

d) Actuator pressure(s) (see note)

e) Valve position

f) Repeat the preceding process for the specified number of points.

b) Dynamic test method

1) Data will be acquired at intervals of time over the duration of the test. The recommended minimum number of data points in each direction is 101 for a full signal span.

2) User defines signal range, ramp, and pause time over which the test is performed.

3) Starting with the user defined minimum signal specified, ramp the input signal to the maximum signal specified, wait the pause time, and ramp down to the minimum signal. Then wait the pause time again.

4) During the travel, record the following data at each data point.

   a) Input signal

   b) Control signal

   c) I/P transducer output, if used

   d) Positioner output or actuator pressure(s)

   e) Valve position

5 Data reporting

5.1 Tagging information

The following is a complete list of data that should be recorded (if available). Include units as applicable.

a) Date and time of test

b) Owner of valve

c) Plant name and location

d) Valve serial number

e) Valve tag number

f) Valve position

g) Valve type

h) Model number

i) Serial number

j) Manufacturer

k) Valve size

l) Inherent flow characteristic

m) Actuator type (spring diaphragm, double acting cylinder, etc.)

n) Actuator manufacturer

o) Air action (air to open, air to close, double acting, etc.)

p) Actuator model number

q) Actuator spring range (not applicable to double acting actuators without spring assisted return)

r) Positioner type (analog or digital)

s) Positioner model number

t) Positioner manufacturer

u) Positioner signal range

v) I/P transducer model number

w) I/P transducer manufacturer

x) I/P signal range

y) Required minimum supply pressure

5.2 Diagnostic test data (see definitions)

The following information is to be recorded at each data point as applicable to the signature.

a) Control signal at each data point, if used to generate test

b) Input signal at each data point

c) I/P transducer output at each data point

d) Positioner output (required for transducer output versus control signal signature and valve position versus actuator pressure signature only). See note.

e) Actuator signal pressure(s) at each data point (required for transducer output versus control signal signature and valve position versus actuator pressure signature only). See note.

NOTE — The actuator pressure, transducer output pressure or positioner output pressure may be assumed the same if the transducer (in cases where no positioner is used) output or electro-pneumatic positioner output is connected directly to the actuator or by an unobstructed conduit.

f) Valve position at each data point

g) Supply pressure at each data point (if available)

h) Ramp time (if available)

i) Pause time (if available)

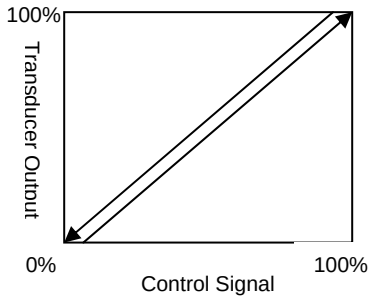
5.3 Graphing

The following graphical representations of signatures will be provided as applicable.

a) Transducer output versus control signal signature

1) Vertical axis - I/P transducer output

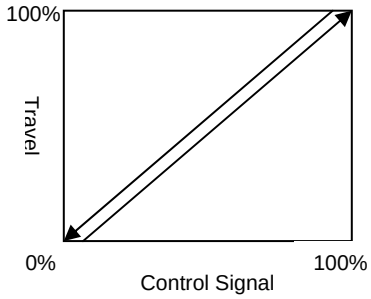
2) Horizontal axis - control signal



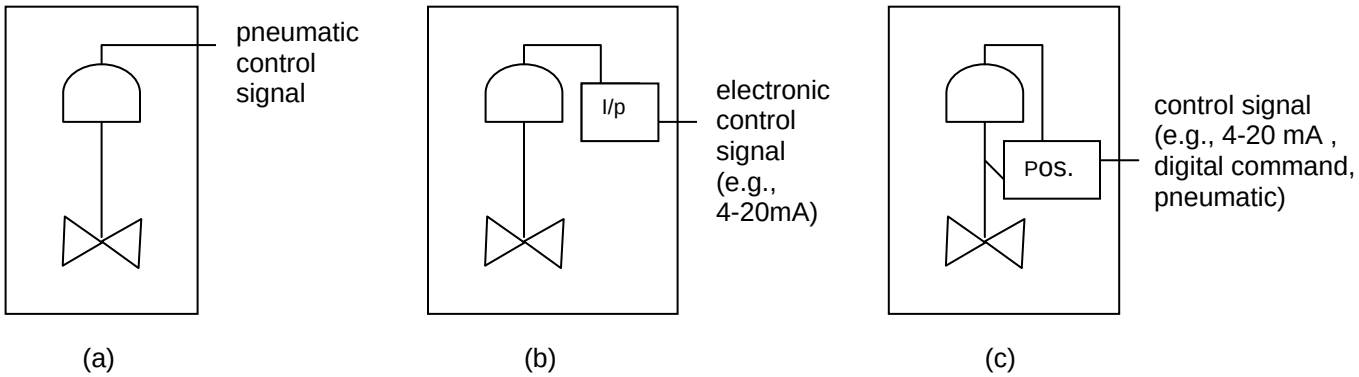
b) Valve position versus control signal signature

1) Vertical axis– travel

2) Horizontal axis – control signal (signal to positioner, control signal direct to actuator or I/P transducer output; see Figures a, b and c below)



Typical examples:

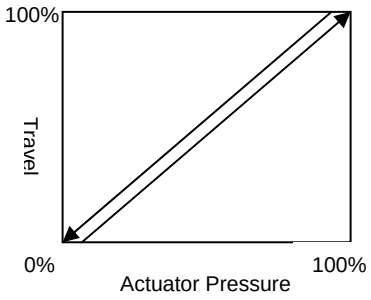


Valve position versus actuator pressure(s) signature

1) Vertical axis - travel

2) Horizontal axis - actuator input pressure(s)

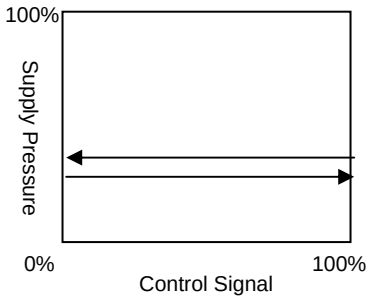
3) Must specify if control signal or actuator pressure was varied to generate the signature.



c) Supply pressure versus control signal signature

1) Vertical axis - supply pressure

2) Horizontal axis - control signal



5.4 Tabulated data

The following data will be provided in an electronic comma delimited format that is compatible with spreadsheet programs (such as Microsoft Excel).

a) Data point number

b) Input signal

c) Control signal

d) Actuator pressure

e) Travel position

f) Ramp time (if available)

g) Pause time (if available)

h) Supply pressure

5.5 Sample file format

```
Filename:,Sample.csv,Date Created:,24 Mar 2004
[Tag Header]
Date and time of test, 24 Mar 2004  04:20:14 pm
Owner of Valve, John Doe Enterprises
(Some lines skipped here)
Rated Valve Travel, 7.5,in
Actual Valve Travel, 7.6,in
Model Number,Mark One
(Some lines skipped here)
[Data Header]
Ramp Time, 10, sec
Pause Time, 20,sec
Point No., Input Signal, Control Signal, Actuator p1, Actuator p2, Travel,
Supply Pressure
(NA), (psig), (psig), (psig), (psig), (in), (psig)
(NA),(+/-0.2), (+/-0.08), (+/-0.1), (+/-0.2), (+/-0.005), (+/-0.5)
[Data]
1, 3.0,3.2,50.0,45.0,0.000,80.0
(Some lines skipped here)
101,16.1,16.2,47.0,44.0,4.500,79.0
```

6 Quality

6.1 All portable data acquisition devices will be calibrated as required by a quality system that is acceptable to the valve owner or operator.

6.2 Valve mounted data acquisition devices must be mounted, calibrated, and tuned to manufacturer specifications to yield accurate valve data.

7 Resolution of measured data

The following are minimum resolution requirements. Some diagnostic data analysis may require greater resolution. Manufacturer specifications may be consulted to determine the actual resolution of the data reported.

7.1 Pressure measurements must be reported to 0.1psi.

7.2 Linear travel must be reported to 0.001 in.

7.3 Rotary valve travel must be reported to 0.1 deg.

7.4 Control signal reporting must be reported to 1% of full span.

7.5 Time must be reported to 1% of total time.

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